

Subject Code	AF5640
Subject Title	Metaverse Economics and Ecosystems
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil (but knowledge in economics is preferable)
Objectives	This subject course introduces the concepts and principles of economics and business ecosystems in applications. The course will provide students with the fundamentals of business analytics for the advanced business architecture of metaverse, covering methods, techniques, systems, and applications. The course will also cover ethical and regulatory aspects of Metaverse economics. The course will equip future AI and metaverse professionals with sound business sense.
Intended Learning Outcomes <i>(Note 1)</i>	Upon completion of the subject, students will be able to: (a) Show in-depth understanding of the fundamental economic concepts, research issues and business challenges, and the state-of-the-art commercial development and applications of Metaverse. (b) Research into the business case studies of the comprehensive business architecture of Metaverse, including understanding the ethical, regulatory, and compliance risks of Metaverse. (c) Learn and deploy the business analytics of a digital, innovative economy, including the advanced business applications of Metaverse and AI. (d) Discover potential Metaverse-oriented business application scenarios and apply business analytics methods and techniques to solve and verify various challenging Metaverse problems for practical applications with innovative solutions.
Subject Synopsis/ Indicative Syllabus <i>(Note 2)</i>	<u>Fundamentals</u> ▪ Introduction to Metaverse: Web 3.0, From AR to XR, Characteristics ▪ The Open Economic System of Metaverse ▪ The Metaverse Economy: Currency, Decentralized Finance, Community Commerce <u>Advanced topics</u> ▪ The “Functionality” of Metaverse: Gaming, Sports, Media and Entertainment, Education, and Power in Virtual Worlds ▪ “Societal Architecture” of Metaverse: The Power of an Immersive Virtual Community

	▪ The Technical “Building Block” of Metaverse ▪ Business architecture of Metaverse: Data, Identify, Ownership in Virtual Worlds ▪ The New Business Model in Metaverse: Experience Marketing ▪ Too “Meta” to fail? Power of Data Vendors and Meta Application Creators ▪ RegTech for Metaverse: The ethical and regulatory dimensions of Metaverse ▪ The Changing World of Metaverse: How real compliance, regulatory, business, and ethical risks are shifting Metaverse <u>Applications</u> ▪ NFT Market ▪ Business Impact of Web3.0 Technology ▪ Consumers in the Metaverse Economy ▪ Business Analytics for New Metaverse Business Opportunities						
Teaching/Learning Methodology <i>(Note 3)</i>	The course is comprised of lectures, tutorials, seminars, and case studies. Especially, the course adopts a case-study approach for teaching, and group projects will be a critical component of the course. During lectures, students are taught the important concepts, principles, and business analytics that drive the development of Metaverse. During tutorials, students will be presented with both theoretical questions and practical scenarios of the Metaverse, and are required to study, analyze and propose solutions. During group projects, small group discussions will be encouraged, and students will need to present their results and solutions in the form of reports and presentations. Students are also required to give seminar talks about topics and case studies that are designed to reinforce their understanding of the knowledge taught and to explore further topics and issues.						
Assessment Methods in Alignment with Intended Learning Outcomes <i>(Note 4)</i>	Specific assessment methods/tasks	% Weighting	Intended subject learning outcomes to be assessed				
			a	b	c	d	e
	1. Classroom participation	5%	✓	✓	✓		
	2. Assignments	25%	✓	✓	✓	✓	

	3. Group projects	35%	✓	✓	✓	✓		
	4. Final Examination	35%	✓	✓	✓	✓		
	Total	100 %						
	<i>Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:</i> Students will be assessed by their performance in four parts: 1) Classroom participation, 2) Assignments, 3) Group projects, and 4) and Examination. - Classroom participation is the mechanism to engage students to actively participate in learning and research by identifying and exploring important issues and problems and presenting existing / promising solutions. - Assignments are written homework that assesses students' abilities to comprehend basic economic concepts and principles, and problem-solving. - Group projects involve a group of students designing and implementing a business solution for a practical Metaverse application. In the group project, students will collaboratively work together to apply what they have learned in the class to solve practical problems. The results are to be presented in the form of reports and / or presentations. - The course will have a final exam.							
Student Study Effort Expected	Class contact:							
	Lecture						26 Hrs.	
	Tutorial / Seminar						13 Hrs.	
	Other student study effort:							
	Self-study: Reading, Discussion, etc.						60 Hrs.	
	Collaborative group project						23 Hrs.	
	Total student study effort						122 Hrs.	
Reading List and References	Reference Books:							
	1. "Metaverse Development: A Step-by-Step Handbook", by Prasanna Bhattathiri (Educohack Press 2025) 2. "Gaming the Metaverse", by Benjamin Beil, Gundolf S. Freyermuth, Isabelle Hamm, Vanessa Ossa (Open Access Publication 2025)							

	3. “The Metaverse and Space Informatics”, by Yung Kai-leung, Andrew Ip Wai-hung, Tang Yuk-ming (PolyU Press 2024). 4. “The Metaverse: And How It Will Revolutionize Everything”, by Matthew Ball (Liveright 2022) 5. “Navigating the Metaverse: A Guide to Limitless Possibilities in a Web 3.0 World”, by Cathy Hackl <i>et al.</i> (Wiley 2022). 6. “The Metaverse Handbook: Innovating for the Internet's Next Tectonic Shift”, by QuHarrison Terry <i>et al.</i> (Wiley 2022). 7. “Step into the Metaverse: How the Immersive Internet Will Unlock a Trillion-Dollar Social Economy”, by Mark van Rijmenam (Wiley 2022). 8. “A Systemic Approach to Continuous Change in the Innovation Economy”, by Jon-Arild Johannessen (Routledge 2021). 9. “The Political Economy of Digital Automation Measuring its Impact on Productivity, Economic Growth, and Consumption”, by Sreenath Majumder and Anuradha SenGupta (Routledge 2021). 10. “The REGTECH Book: The Financial Technology Handbook for Investors, Entrepreneurs and Visionaries in Regulation”, by Janos Barberis <i>et al.</i> (Wiley 2019). 11. “Snow Crash”, by Neal Stephenson (Spectra 2003)
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Note 1: Intended Learning Outcomes

Intended learning outcomes should state what students should be able to do or attain upon subject completion. Subject outcomes are expected to contribute to the attainment of the overall programme outcomes.

Note 2: Subject Synopsis/Indicative Syllabus

The syllabus should adequately address the intended learning outcomes. At the same time, overcrowding of the syllabus should be avoided.

Note 3: Teaching/Learning Methodology

This section should include a brief description of the teaching and learning methods to be employed to facilitate learning, and a justification of how the methods are aligned with the intended learning outcomes of the subject.

Note 4: Assessment Method

This section should include the assessment method(s) to be used and its relative weighting, and indicate which of the subject intended learning outcomes that each method is intended to assess. It should also provide a brief explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes.

(Form AR 140) 8.2020